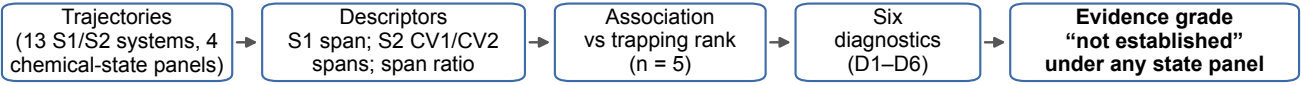

Six diagnostics → outcomes in this study (n = 5, tie-aware exact permutation)

D1 Windowed estimator variance	variability vs separations (2.6–5.3 kcal/mol); no stable order at 26 ns
D2 Estimator–consistency matrix	no first-window cell provides evidence (all $p \geq 0.133$, 48 cells); sign unstable
D3 Convergence–timescale scaling	no decay over 2–24 ns windows; no extrapolated sampling length
D4 Ordinal sensitivity model	S1 favors positive at narrowest band ($P(\rho<0)=0.01$); band-dependent
D5 Counterfactual sensitivity	leave-one-out −0.63…+0.63; single-point exclusion flips sign
D6 Chemical-state audit	state corrections move ρ (−0.15…+0.36); state vs noise unresolved

Verdict: no statistically supported evidence of a robust span–trapping association; not established under any chemically defensible ligand-state panel. The output of this work is the audit workflow itself.